CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

FANELLI DAIRY

7689 WASHINGTON RD

HILMAR CA 95324

NOTES:

PREPARED BY:

CARDOSO AG SERVICES INC.
PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Fanelli Dairy

Physical address of dairy:

7689 Washington RD	Hillmar	Merced	95324
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1971

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X190-X007-XXXX

B. OPERATORS

Fanelli, Victor

Operator name: Fanelli, Victor

Telephone no.: (209) 668-8097 Landline (209) 678-2858 Cellular

7689 Washington RD	Hilmar	CA	95324
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Fanelli, Victor

Legal owner name: Fanelli, Victor

Telephone no.: (209) 668-8097 Landline (209) 678-2858 Cellular

7689 Washington RD	Hilmar	CA	95324
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	615	130	432	0	0	0
Maximum number	615	130	432	0	0	0
Average number	615	130	432	0	0	0
Avg live weight (lbs)	1,000	1,050	725	0		

Predominant milk cow breed: Jersey

Average milk production: 51 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 20,212.20 *tons per reporting period*

Total nitrogen from manure: 253,050.40 *lbs per reporting period*

After ammonia losses (30% loss applied): 177,135.28 *lbs per reporting period*

Total phosphorus from manure: 41,691.12 *lbs per reporting period*

Total potassium from manure: 110,101.13 *lbs per reporting period*

Total salt from manure: 319,466.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 7,798,964 *gallons*

Total nitrogen generated: 36,380.36 *lbs*

Total phosphorus generated: 6,571.21 *lbs*

Total potassium generated: 58,750.78 *lbs*

Total salt generated: 251,061.87 *lbs*

	7,798,964	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	7,798,964	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
01/01/2023	Separator solids	288.00 *ton*	Dry-weight	74.1		16,400.00	3,000.00	4,300.00		14.30
04/29/2023	Separator solids	670.00 *ton*	Dry-weight	74.1		16,400.00	3,000.00	4,300.00		14.30

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	8,138.40	1,488.73	2,133.85	70,962.89
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	8,138.40	1,488.73	2,133.85	70,962.89

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	17	17	2	both	X045-X190-X007-XXXX
2	20	20	2	both	X045-X190-X008-XXXX
3	18	18	2	process wastewater	X045-X190-X008-XXXX
Carvalho North	15	15	2	process wastewater	X045-X190-X075-XXXX X045-X190-X076-XXXX
Carvalho South	15	15	2	process wastewater	X045-X190-X075-XXXX X045-X190-X076-XXXX
Dads	30	30	2	manure	X045-X190-X027-XXXX
Totals for areas that were used for application	115	115	12		
Totals for areas that were not used for application					
Land application area totals	115	115	12		

B. CROPS AND HARVESTS

1

Field name: 1

09/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 17 Plant date: 09/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	251.00 *ton*	Dry-weight		63.7	27,000.00	5,000.00	50,800.00		10.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.76	289.42	53.60	544.53	1,146.95

1

05/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 05/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	382.00 *ton*	Dry-weight		65.0	15,200.00	3,400.00	22,300.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	22.47	239.09	53.48	350.77	1,211.16

2

Field name: 2

09/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 09/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	295.00 *ton*	Dry-weight		63.7	27,000.00	5,000.00	50,800.00		10.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.75	289.13	53.54	543.99	1,145.81

05/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 05/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	450.00 *ton*	Dry-weight		65.0	15,200.00	3,400.00	22,300.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	22.50	239.40	53.55	351.23	1,212.75

3

Field name: 3

09/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 09/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	265.00 *ton*	Dry-weight		75.0	27,600.00	4,700.00	48,200.00		11.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.72	203.17	34.60	354.81	846.53

05/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 05/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	405.00 *ton*	Dry-weight		57.9	12,900.00	3,000.00	22,900.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	22.50	244.39	56.84	433.84	1,307.21

Carvalho North

Field name: Carvalho North

09/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 09/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	221.00 *ton*	Dry-weight		62.3	16,300.00	4,100.00	44,900.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.73	181.08	45.55	498.79	944.26

Carvalho North

05/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 05/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	337.00 *ton*	Dry-weight		66.2	12,400.00	3,100.00	19,400.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	22.47	188.32	47.08	294.64	956.81

Carvalho South

Field name: Carvalho South

09/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 09/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	221.00 *ton*	Dry-weight		77.7	21,200.00	2,700.00	15,100.00		11.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.73	139.31	17.74	99.22	735.96

05/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 05/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	337.00 *ton*	Dry-weight		62.0	12,000.00	1,700.00	20,400.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	22.47	204.90	29.03	348.32	956.18

Dads

Field name: Dads

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	442.00 *ton*	Dry-weight		58.0	13,800.00	2,100.00	12,400.00		5.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.73	170.79	25.99	153.46	680.68

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	675.00 *ton*	Dry-weight		70.5	13,600.00	2,800.00	14,600.00		4.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	22.50	180.54	37.17	193.82	637.20

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 09/10/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 09/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	111.94	20.48	29.35	976.03	224.00 *ton*
Application event totals		111.94	20.48	29.35	976.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	81.31	11.83	139.01	645.66	264,649.00 *gal*
Canal	Surface water	16.26	0.00	0.00	0.00	5,174,542.00 *gal*
Application event totals		97.57	11.83	139.01	645.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	92.17	13.92	152.00	626.28	286,703.00 *gal*
Application event totals		92.17	13.92	152.00	626.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	92.17	13.92	152.00	626.28	286,703.00 *gal*
Application event totals		92.17	13.92	152.00	626.28	

1 - 05/17/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 05/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

1 - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	70.90	10.71	116.92	481.76	220,541.00 *gal*
Canal	Surface water	17.63	0.00	0.00	0.00	5,613,008.00 *gal*
Application event totals		88.53	10.71	116.92	481.76	

06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	70.90	10.71	116.92	481.76	220,541.00 *gal*
Canal	Surface water	8.82	0.00	0.00	0.00	2,806,520.00 *gal*
Application event totals		79.72	10.71	116.92	481.76	

06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.17	0.00	0.00	0.00	1,964,558.00 *gal*
Application event totals		6.17	0.00	0.00	0.00	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	7.05	0.00	0.00	0.00	2,245,223.00 *gal*
Application event totals		7.05	0.00	0.00	0.00	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.29	0.00	0.00	0.00	1,683,893.00 *gal*
Application event totals		5.29	0.00	0.00	0.00	

1 - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	53.30	15.34	77.12	337.77	286,703.00 *gal*
Canal	Surface water	7.05	0.00	0.00	0.00	2,245,223.00 *gal*
Application event totals		60.35	15.34	77.12	337.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.17	0.00	0.00	0.00	1,964,558.00 *gal*
Application event totals		6.17	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	53.30	15.34	77.12	337.77	286,703.00 *gal*
Canal	Surface water	4.41	0.00	0.00	0.00	1,403,260.00 *gal*
Application event totals		57.71	15.34	77.12	337.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.29	0.00	0.00	0.00	1,683,893.00 *gal*
Application event totals		5.29	0.00	0.00	0.00	

2 - 09/10/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough

Plant date: 09/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

2 - 09/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	112.14	20.51	29.40	977.78	264.00 *ton*
Application event totals		112.14	20.51	29.40	977.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	81.31	11.83	139.00	645.66	311,351.00 *gal*
Canal	Surface water	17.27	0.00	0.00	0.00	6,465,925.00 *gal*
Application event totals		98.58	11.83	139.00	645.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	92.17	13.92	152.00	626.28	337,297.00 *gal*
Application event totals		92.17	13.92	152.00	626.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	92.17	13.92	152.00	626.28	337,297.00 *gal*
Application event totals		92.17	13.92	152.00	626.28	

2 - 05/17/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 05/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

2 - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	70.90	10.71	116.92	481.75	259,459.00 *gal*
Canal	Surface water	7.39	0.00	0.00	0.00	2,765,656.00 *gal*
Application event totals		78.28	10.71	116.92	481.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	70.90	10.71	116.92	481.75	259,459.00 *gal*
Canal	Surface water	12.37	0.00	0.00	0.00	4,632,823.00 *gal*
Application event totals		83.27	10.71	116.92	481.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.25	0.00	0.00	0.00	3,839,136.00 *gal*
Application event totals		10.25	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.03	0.00	0.00	0.00	3,380,321.00 *gal*
Application event totals		9.03	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	7.18	0.00	0.00	0.00	2,689,910.00 *gal*
Application event totals		7.18	0.00	0.00	0.00	

2 - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	53.30	15.34	77.12	337.77	337,297.00 *gal*
Canal	Surface water	7.74	0.00	0.00	0.00	2,898,246.00 *gal*
Application event totals		61.04	15.34	77.12	337.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	7.74	0.00	0.00	0.00	2,898,246.00 *gal*
Application event totals		7.74	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	53.30	15.34	77.12	337.77	337,297.00 *gal*
Canal	Surface water	6.44	0.00	0.00	0.00	2,410,236.00 *gal*
Application event totals		59.73	15.34	77.12	337.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	7.61	0.00	0.00	0.00	2,850,596.00 *gal*
Application event totals		7.61	0.00	0.00	0.00	

3 - 09/10/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough

Plant date: 09/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

3 - 09/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	83.57	12.16	142.87	663.59	288,000.00 *gal*
Canal	Surface water	17.27	0.00	0.00	0.00	5,819,332.00 *gal*
Application event totals		100.84	12.16	142.87	663.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	87.44	13.21	144.20	594.16	288,000.00 *gal*
Application event totals		87.44	13.21	144.20	594.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	87.44	13.21	144.20	594.16	288,000.00 *gal*
Application event totals		87.44	13.21	144.20	594.16	

3 - 05/17/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 05/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	70.90	10.71	116.92	481.76	233,514.00 *gal*
Canal	Surface water	7.39	0.00	0.00	0.00	2,489,090.00 *gal*
Application event totals		78.28	10.71	116.92	481.76	

3 - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	70.90	10.71	116.92	481.76	233,514.00 *gal*
Canal	Surface water	12.37	0.00	0.00	0.00	4,169,541.00 *gal*
Application event totals		83.27	10.71	116.92	481.76	

06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.25	0.00	0.00	0.00	3,455,223.00 *gal*
Application event totals		10.25	0.00	0.00	0.00	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.03	0.00	0.00	0.00	3,042,289.00 *gal*
Application event totals		9.03	0.00	0.00	0.00	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	53.30	15.34	77.12	337.77	303,568.00 *gal*
Canal	Surface water	7.18	0.00	0.00	0.00	2,420,919.00 *gal*
Application event totals		60.48	15.34	77.12	337.77	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	7.74	0.00	0.00	0.00	2,608,421.00 *gal*
Application event totals		7.74	0.00	0.00	0.00	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	7.74	0.00	0.00	0.00	2,608,421.00 *gal*
Application event totals		7.74	0.00	0.00	0.00	

3 - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.44	0.00	0.00	0.00	2,169,213.00 *gal*
Application event totals		6.44	0.00	0.00	0.00	

08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	53.30	15.34	77.12	337.77	303,568.00 *gal*
Canal	Surface water	7.61	0.00	0.00	0.00	2,565,536.00 *gal*
Application event totals		60.91	15.34	77.12	337.77	

Carvalho North - 09/10/2022: Oats, silage-soft dough

Field name: Carvalho North

Crop: Oats, silage-soft dough Plant date: 09/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	86.08	12.53	147.15	683.50	247,200.00 *gal*
Canal	Surface water	16.30	0.00	0.00	0.00	4,578,895.00 *gal*
Application event totals		102.38	12.53	147.15	683.50	

03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	72.58	10.96	119.69	493.16	199,200.00 *gal*
Application event totals		72.58	10.96	119.69	493.16	

Carvalho North - 09/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	70.83	10.70	116.80	481.27	194,400.00 *gal*
Application event totals		70.83	10.70	116.80	481.27	

Carvalho North - 05/19/2023: Corn, silage

Field name: Carvalho North

Crop: Corn, silage

Plant date: 05/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	56.84	8.58	93.73	386.21	156,000.00 *gal*
Canal	Surface water	13.33	0.00	0.00	0.00	3,743,516.00 *gal*
Application event totals		70.17	8.58	93.73	386.21	

06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.45	0.00	0.00	0.00	2,654,358.00 *gal*
Application event totals		9.45	0.00	0.00	0.00	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	56.84	8.58	93.73	386.21	156,000.00 *gal*
Canal	Surface water	8.48	0.00	0.00	0.00	2,382,215.00 *gal*
Application event totals		65.32	8.58	93.73	386.21	

Carvalho North - 05/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.21	0.00	0.00	0.00	2,586,473.00 *gal*
Application event totals		9.21	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.39	10.19	51.22	224.31	168,000.00 *gal*
Canal	Surface water	10.66	0.00	0.00	0.00	2,995,282.00 *gal*
Application event totals		46.06	10.19	51.22	224.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.21	0.00	0.00	0.00	2,586,473.00 *gal*
Application event totals		9.21	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.39	10.19	51.22	224.31	168,000.00 *gal*
Canal	Surface water	9.45	0.00	0.00	0.00	2,654,358.00 *gal*
Application event totals		44.85	10.19	51.22	224.31	

Carvalho South - 09/10/2022: Oats, silage-soft dough

Field name: Carvalho South

Crop: Oats, silage-soft dough

Plant date: 09/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Carvalho South - 09/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	54.32	7.91	92.86	431.34	156,000.00 *gal*
Canal	Surface water	16.30	0.00	0.00	0.00	4,578,895.00 *gal*
Application event totals		70.62	7.91	92.86	431.34	

Carvalho South - 05/19/2023: Corn, silage

Field name: Carvalho South

Crop: Corn, silage

Plant date: 05/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	13.33	0.00	0.00	0.00	3,743,516.00 *gal*
Application event totals		13.33	0.00	0.00	0.00	

06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.45	0.00	0.00	0.00	2,654,358.00 *gal*
Application event totals		9.45	0.00	0.00	0.00	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.48	0.00	0.00	0.00	2,382,215.00 *gal*
Application event totals		8.48	0.00	0.00	0.00	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.21	0.00	0.00	0.00	2,586,473.00 *gal*
Application event totals		9.21	0.00	0.00	0.00	

Carvalho South - 05/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	40.45	11.64	58.54	256.36	192,000.00 *gal*
Canal	Surface water	10.66	0.00	0.00	0.00	2,995,282.00 *gal*
Application event totals		51.12	11.64	58.54	256.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.21	0.00	0.00	0.00	2,586,473.00 *gal*
Application event totals		9.21	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	40.45	11.64	58.54	256.36	192,000.00 *gal*
Canal	Surface water	9.45	0.00	0.00	0.00	2,654,358.00 *gal*
Application event totals		49.90	11.64	58.54	256.36	

Dads - 11/05/2022: Oats, silage-soft dough

Field name: Dads

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	222.57	40.71	58.36	1,940.74	786.00 *ton*
Application event totals		222.57	40.71	58.36	1,940.74	

Dads - 11/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.15	0.00	0.00	0.00	5,703,771.00 *gal*
Application event totals		10.15	0.00	0.00	0.00	

Dads - 06/03/2023: Corn, silage

Field name: Dads

Crop: Corn, silage

Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Separator solids	169.26	45.75	66.79	0.00	267.00 *ton*
Application event totals		169.26	45.75	66.79	0.00	

05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.14	0.00	0.00	0.00	1,199,964.00 *gal*
Application event totals		2.14	0.00	0.00	0.00	

06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.34	0.00	0.00	0.00	3,561,794.00 *gal*
Application event totals		6.34	0.00	0.00	0.00	

06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.29	0.00	0.00	0.00	3,534,125.00 *gal*
Application event totals		6.29	0.00	0.00	0.00	

Dads - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	14.09	0.00	0.00	0.00	7,912,735.00 *gal*
Application event totals		14.09	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.67	0.00	0.00	0.00	6,556,554.00 *gal*
Application event totals		11.67	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	12.23	0.00	0.00	0.00	6,867,756.00 *gal*
Application event totals		12.23	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.90	0.00	0.00	0.00	6,120,125.00 *gal*
Application event totals		10.90	0.00	0.00	0.00	

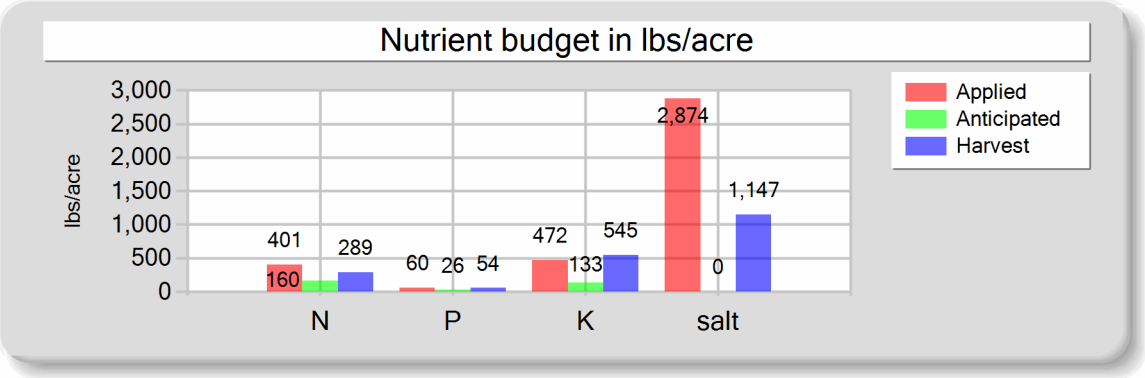
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.76	0.00	0.00	0.00	6,607,003.00 *gal*
Application event totals		11.76	0.00	0.00	0.00	

B. NUTRIENT BUDGET

1 - 09/10/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 09/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	111.94	20.48	29.35	976.03
Process wastewater	265.65	39.67	443.00	1,898.22
Fresh water	16.26	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	400.84	60.15	472.35	2,874.26
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	289.42	53.60	544.53	1,146.95
Nutrient balance	111.43	6.55	-72.19	1,727.31
Applied to removed ratio	1.38	1.12	0.87	2.51

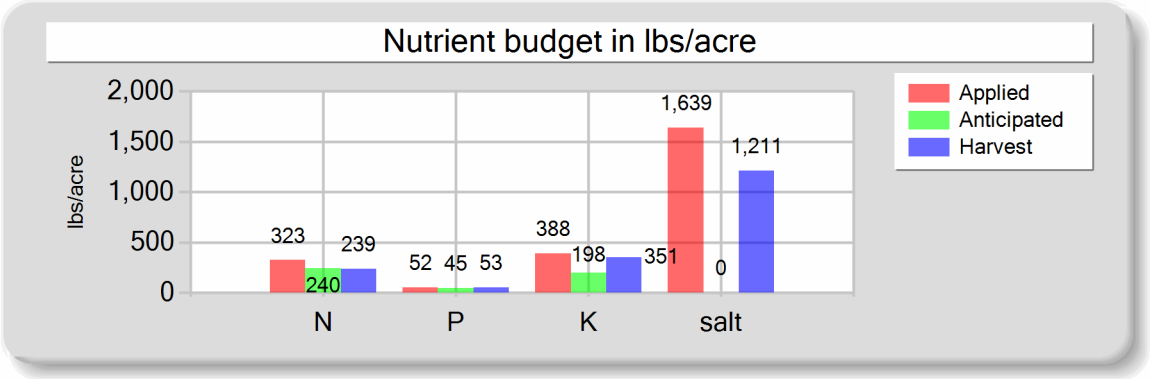
Fresh water applied	
5,174,542.00	*gallons*
190.56	*acre-inches*
11.21	*inches/acre*

Process wastewater applied	
838,055.00	*gallons*
30.86	*acre-inches*
1.82	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 05/17/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 05/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	248.39	52.09	388.09	1,639.05
Fresh water	67.89	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	323.28	52.09	388.09	1,639.05
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	239.09	53.48	350.77	1,211.16
Nutrient balance	84.20	-1.39	37.32	427.89
Applied to removed ratio	1.35	0.97	1.11	1.35

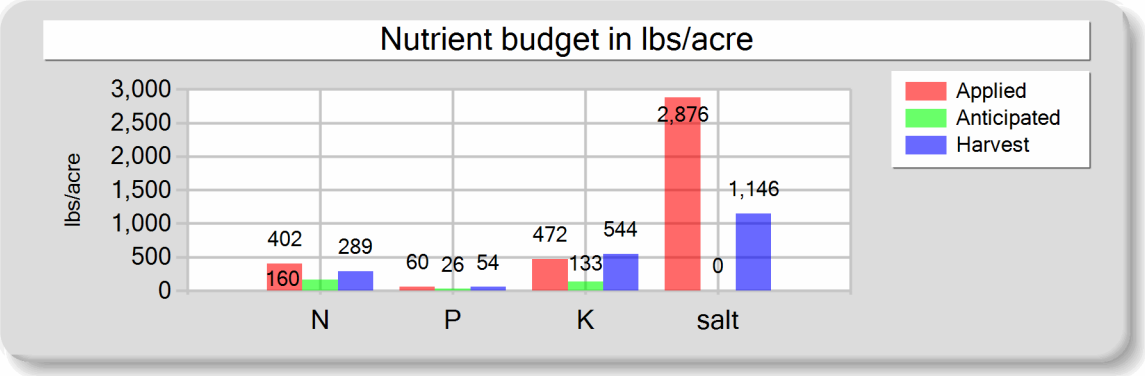
Fresh water applied	
21,610,136.00	*gallons*
795.83	*acre-inches*
46.81	*inches/acre*

Process wastewater applied	
1,014,488.00	*gallons*
37.36	*acre-inches*
2.20	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 09/10/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 09/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	112.14	20.51	29.40	977.78
Process wastewater	265.65	39.67	443.00	1,898.22
Fresh water	17.27	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	402.05	60.19	472.40	2,876.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	289.13	53.54	543.99	1,145.81
Nutrient balance	112.92	6.64	-71.59	1,730.19
Applied to removed ratio	1.39	1.12	0.87	2.51

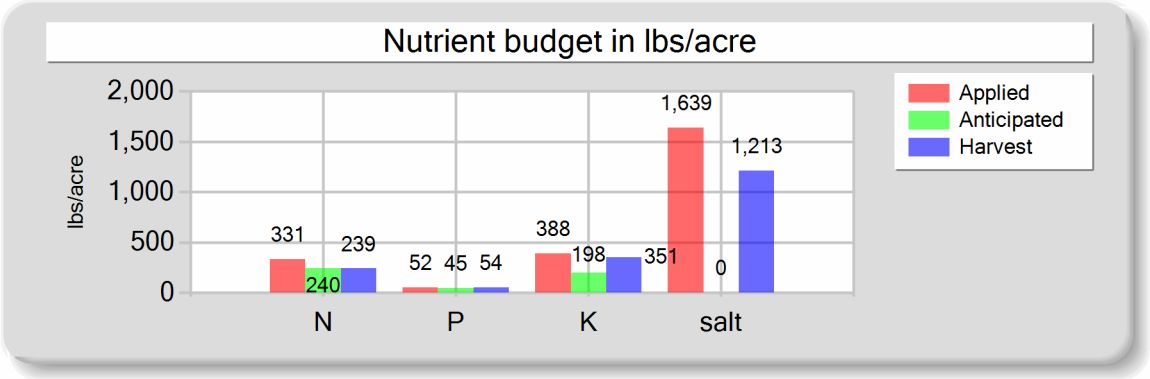
Fresh water applied	
6,465,925.00	*gallons*
238.12	*acre-inches*
11.91	*inches/acre*

Process wastewater applied	
985,945.00	*gallons*
36.31	*acre-inches*
1.82	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 05/17/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 05/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	248.39	52.09	388.09	1,639.05
Fresh water	75.75	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	331.14	52.09	388.09	1,639.05
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	239.40	53.55	351.23	1,212.75
Nutrient balance	91.74	-1.46	36.86	426.30
Applied to removed ratio	1.38	0.97	1.10	1.35

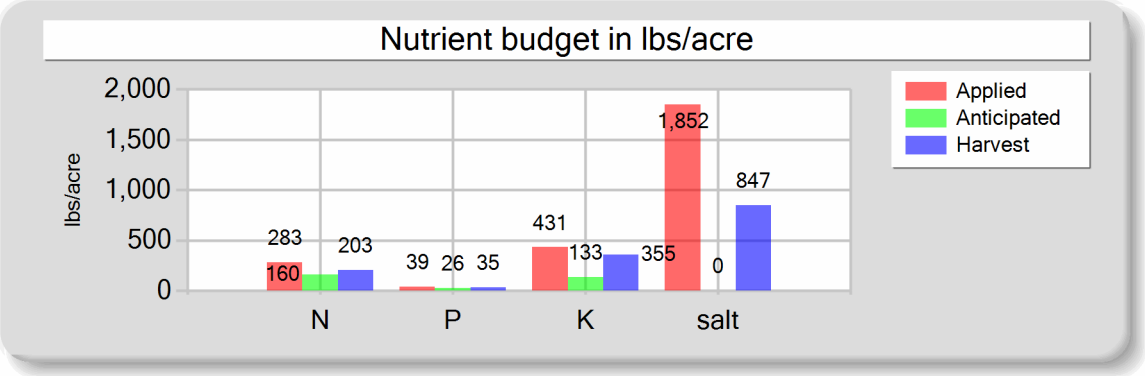
Fresh water applied	
28,365,170.00	*gallons*
1,044.59	*acre-inches*
52.23	*inches/acre*

Process wastewater applied	
1,193,512.00	*gallons*
43.95	*acre-inches*
2.20	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 09/10/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 09/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	258.45	38.57	431.27	1,851.92
Fresh water	17.27	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	282.72	38.57	431.27	1,851.92
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	203.17	34.60	354.81	846.53
Nutrient balance	79.55	3.98	76.46	1,005.39
Applied to removed ratio	1.39	1.11	1.22	2.19

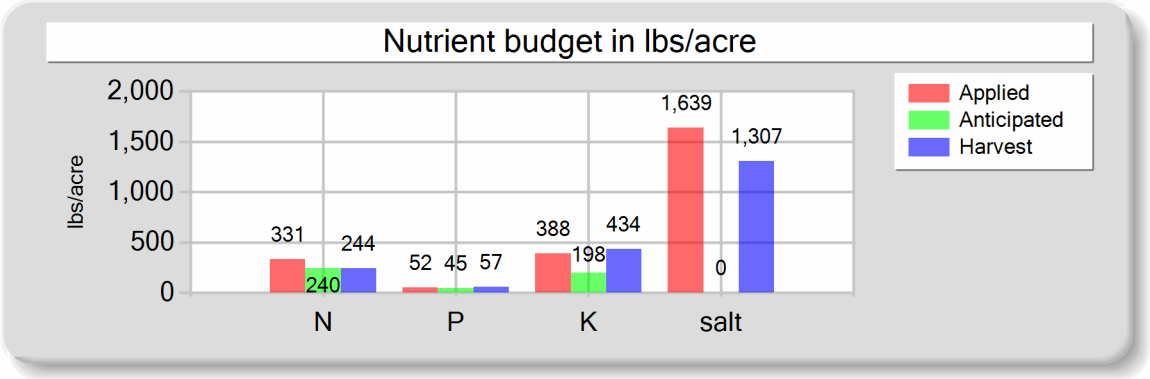
Fresh water applied	
5,819,332.00	*gallons*
214.31	*acre-inches*
11.91	*inches/acre*

Process wastewater applied	
864,000.00	*gallons*
31.82	*acre-inches*
1.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 05/17/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 05/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	248.39	52.09	388.09	1,639.05
Fresh water	75.75	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	331.14	52.09	388.09	1,639.05
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	244.39	56.84	433.84	1,307.21
Nutrient balance	86.75	-4.74	-45.75	331.85
Applied to removed ratio	1.35	0.92	0.89	1.25

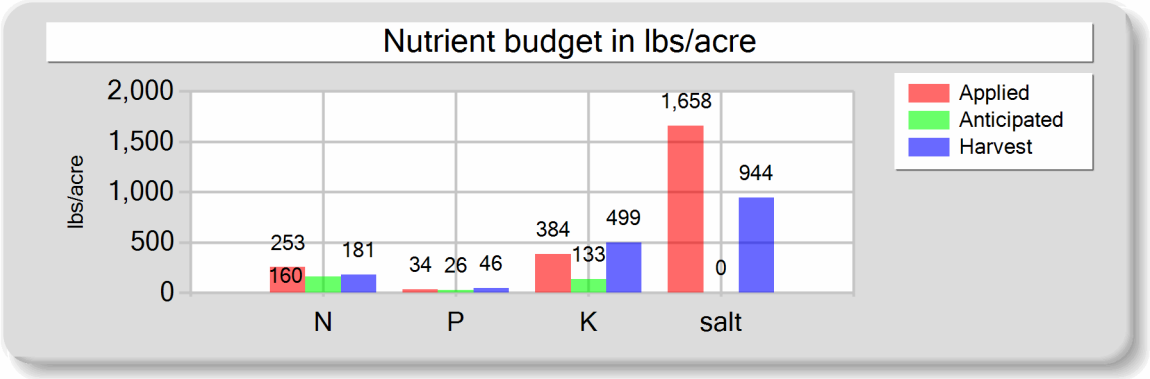
Fresh water applied	
25,528,653.00	*gallons*
940.13	*acre-inches*
52.23	*inches/acre*

Process wastewater applied	
1,074,164.00	*gallons*
39.56	*acre-inches*
2.20	*inches/acre*

Total harvests for the crop	
1	*harvests*

Carvalho North - 09/10/2022: Oats, silage-soft dough

Field name: Carvalho North | Crop: Oats, silage-soft dough | Plant date: 09/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	229.48	34.18	383.64	1,657.93
Fresh water	16.30	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	252.79	34.18	383.64	1,657.93
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	181.08	45.55	498.79	944.26
Nutrient balance	71.71	-11.36	-115.15	713.67
Applied to removed ratio	1.40	0.75	0.77	1.76

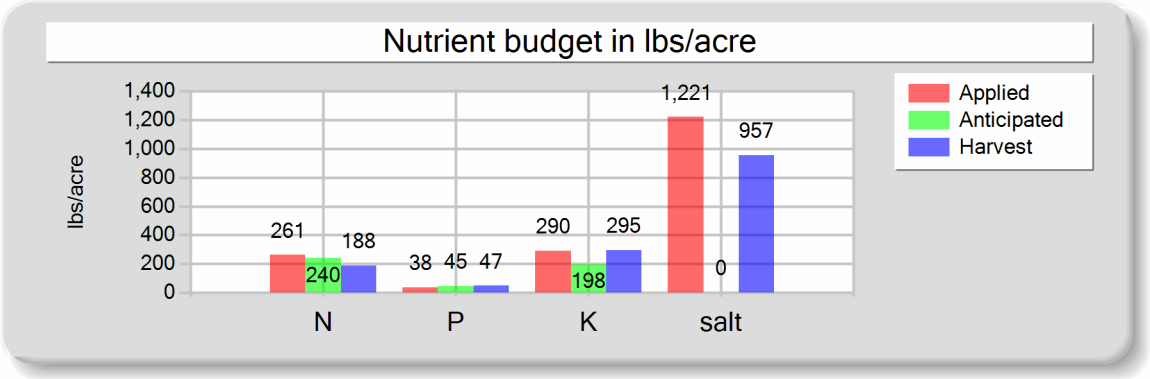
Fresh water applied	
4,578,895.00	*gallons*
168.63	*acre-inches*
11.24	*inches/acre*

Process wastewater applied	
640,800.00	*gallons*
23.60	*acre-inches*
1.57	*inches/acre*

Total harvests for the crop	
1	*harvests*

Carvalho North - 05/19/2023: Corn, silage

Field name: Carvalho North | Crop: Corn, silage | Plant date: 05/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	184.46	37.54	289.90	1,221.04
Fresh water	69.80	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	261.26	37.54	289.90	1,221.04
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	188.32	47.08	294.64	956.81
Nutrient balance	72.94	-9.54	-4.74	264.23
Applied to removed ratio	1.39	0.80	0.98	1.28

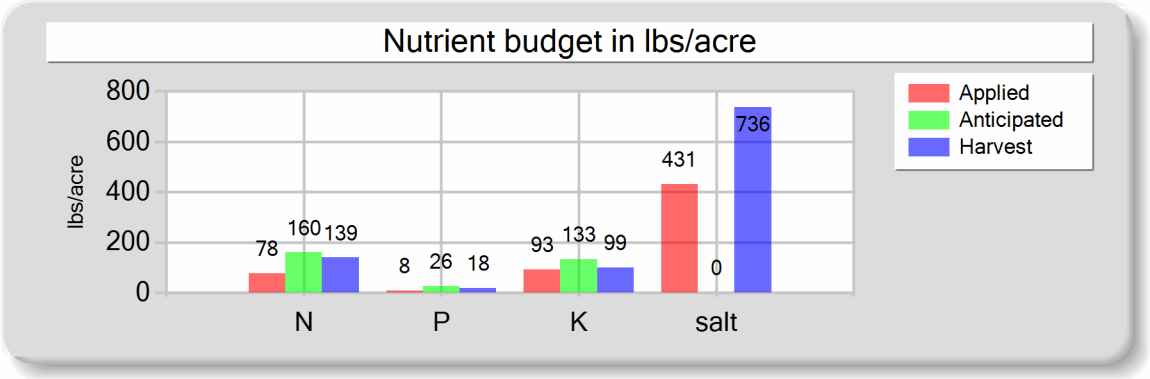
Fresh water applied	
19,602,675.00	*gallons*
721.90	*acre-inches*
48.13	*inches/acre*

Process wastewater applied	
648,000.00	*gallons*
23.86	*acre-inches*
1.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

Carvalho South - 09/10/2022: Oats, silage-soft dough

Field name: Carvalho South | Crop: Oats, silage-soft dough | Plant date: 09/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	54.32	7.91	92.86	431.34
Fresh water	16.30	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	77.62	7.91	92.86	431.34
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	139.31	17.74	99.22	735.96
Nutrient balance	-61.68	-9.84	-6.36	-304.62
Applied to removed ratio	0.56	0.45	0.94	0.59

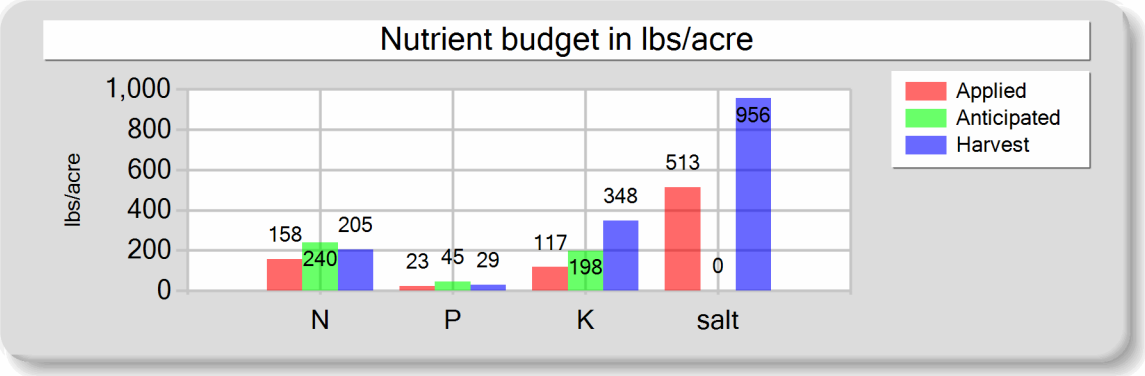
Fresh water applied	
4,578,895.00	*gallons*
168.63	*acre-inches*
11.24	*inches/acre*

Process wastewater applied	
156,000.00	*gallons*
5.74	*acre-inches*
0.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

Carvalho South - 05/19/2023: Corn, silage

Field name: Carvalho South | Crop: Corn, silage | Plant date: 05/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	80.90	23.29	117.07	512.72
Fresh water	69.80	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	157.70	23.29	117.07	512.72
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	204.90	29.03	348.32	956.18
Nutrient balance	-47.20	-5.74	-231.25	-443.46
Applied to removed ratio	0.77	0.80	0.34	0.54

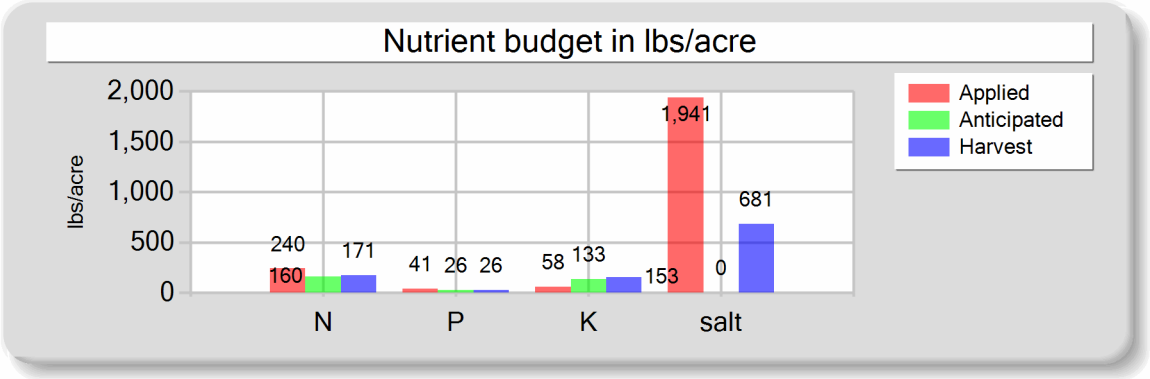
Fresh water applied	
19,602,675.00	*gallons*
721.90	*acre-inches*
48.13	*inches/acre*

Process wastewater applied	
384,000.00	*gallons*
14.14	*acre-inches*
0.94	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dads - 11/05/2022: Oats, silage-soft dough

Field name: Dads | Crop: Oats, silage-soft dough | Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	222.57	40.71	58.36	1,940.74
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	10.15	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	239.73	40.71	58.36	1,940.74
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	170.79	25.99	153.46	680.68
Nutrient balance	68.94	14.73	-95.10	1,260.06
Applied to removed ratio	1.40	1.57	0.38	2.85

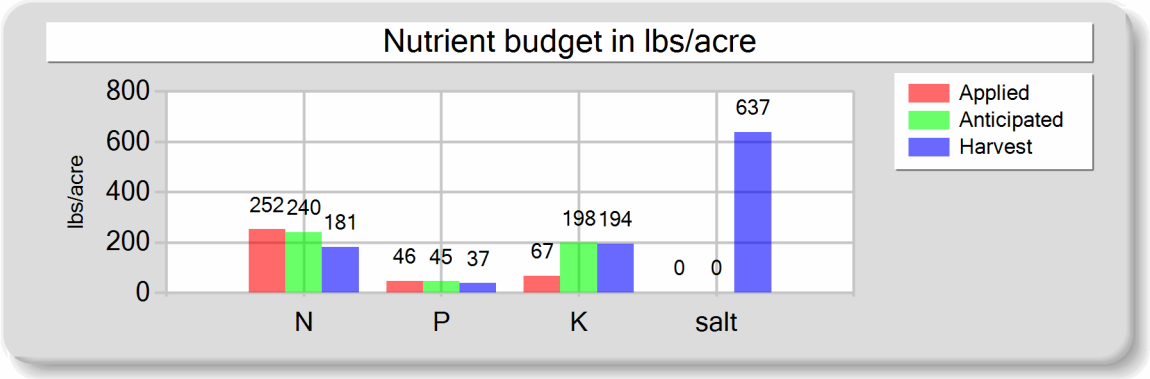
Fresh water applied	
5,703,771.00	*gallons*
210.05	*acre-inches*
7.00	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dads - 06/03/2023: Corn, silage

Field name: Dads | Crop: Corn, silage | Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	169.26	45.75	66.79	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	75.41	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	251.67	45.75	66.79	0.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	180.54	37.17	193.82	637.20
Nutrient balance	71.13	8.58	-127.03	-637.20
Applied to removed ratio	1.39	1.23	0.34	0.00

Fresh water applied	
42,360,056.00	*gallons*
1,559.98	*acre-inches*
52.00	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 Separator Manure

Sample and source description: 2022 Separator Manure

Sample date: 05/11/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,400.00	3,000.00	4,300.00							14.30
DL	0.01	0.01	0.01							1.00

Corral Manure

Sample and source description: Corral Manure

Sample date: 05/08/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 31.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,000.00	4,300.00	15,800.00							0.00
DL	0.01	0.01	0.01							1.00

Manure

Sample and source description: Manure

Sample date: 05/10/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 1.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	1.00	1.00	1.00							1.00
DL	1.00	1.00	1.00							1.00

Separator Manure

Sample and source description: Separator Manure

Sample date: 10/05/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 48.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,500.00	5,000.00	7,300.00							0.00
DL	0.01	0.01	0.01							1.00

B. PROCESS WASTEWATER ANALYSES

2022 Lagoon Q4

Sample and source description: 2022 Lagoon Q4

Sample date: 11/03/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	624.00	416.00	0.00	1.90	91.10	1,070.00								6,070.00	4,970
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,300.00	458.00	0.00	1.70	342.00	882.00								8,800.00	2,800
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/09/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	654.00	418.00	0.00	0.90	98.90	1,080.00								9,520.00	4,450
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	378.00	320.00	0.00	0.70	109.00	548.00								6,060.00	2,400
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 11/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	642.00	417.00	0.00	0.90	88.70	744.00								6,070.00	4,970
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

C. FRESH WATER ANALYSES

Canal

Canal

TID Canal

Sample description: TID Canal

Sample date: 09/20/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.40	0.00	6.40								284.00	0
DL	0.01	0.01	0.01								10.00	1

Domestic Well

DW MB

Sample description: DW MB

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	15.30	0.00	15.30								1,080.00	0
DL	0.01	0.01	0.01								10.00	1

DW Cow Pens

Sample description: DW Cow Pens

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	17.70	0.00	17.70								1,080.00	0
DL	0.01	0.01	0.01								10.00	1

D. SOIL ANALYSES

2

2

Field 1 Soil

Sample and source description: Field 1 Soil

Sample date: 09/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			27.00				
DL			0.01				

Field 2 Soil

Sample and source description: Field 2 Soil

Sample date: 09/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			27.00				
DL			0.01				

3

Field 3 Soil

Sample and source description: Field 3 Soil

Sample date: 09/07/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			24.00				
DL			0.01				

Carvalho North

Carvalho North

Carvalho North Soil

Sample and source description: Carvalho North Soil

Sample date: 09/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			28.00				
DL			0.01				

Carvalho South

Carvalho South Soil

Sample and source description: Carvalho South Soil

Sample date: 09/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			28.00				
DL			0.01				

Dads

Dads Field Soil

Sample and source description: Dads Field Soil

Sample date: 09/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			25.00				
DL			0.01				

E. PLANT TISSUE ANALYSES

1 - 09/10/2022: Oats, silage-soft dough

1 - 09/10/2022: Oats, silage-soft dough

Field 1 Oats

Sample and source description: Field 1 Oats

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,000.00	5,000.00	50,800.00		10.70
DL	0.01	0.01	0.01		1.00

1 - 05/17/2023: Corn, silage

Field 1 Corn

Sample and source description: Field 1 Corn

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,200.00	3,400.00	22,300.00		7.70
DL	0.01	0.01	0.01		1.00

2 - 09/10/2022: Oats, silage-soft dough

Field 2 Oats

Sample and source description: Field 2 Oats

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,000.00	5,000.00	50,800.00		10.70
DL	0.01	0.01	0.01		1.00

2 - 05/17/2023: Corn, silage

Field 2 Corn

Sample and source description: Field 2 Corn

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,200.00	3,400.00	22,300.00		7.70
DL	0.01	0.01	0.01		1.00

3 - 09/10/2022: Oats, silage-soft dough

Field 3 Oats

Sample and source description: Field 3 Oats

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,600.00	4,700.00	48,200.00		11.50
DL	0.01	0.01	0.01		1.00

3 - 05/17/2023: Corn, silage

Field 3 Corn

Sample and source description: Field 3 Corn

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,900.00	3,000.00	22,900.00		6.90
DL	0.01	0.01	0.01		1.00

Carvalho North - 09/10/2022: Oats, silage-soft dough

Carvalho North Oats

Sample and source description: Carvalho North Oats

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,300.00	4,100.00	44,900.00		8.50
DL	0.01	0.01	0.01		1.00

Carvalho North - 05/19/2023: Corn, silage

Carvalho North Corn

Sample and source description: Carvalho North Corn

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,400.00	3,100.00	19,400.00		6.30
DL	0.01	0.01	0.01		1.00

Carvalho South - 09/10/2022: Oats, silage-soft dough

Carvalho South Oats

Sample and source description: Carvalho South Oats

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,200.00	2,700.00	15,100.00		11.20
DL	0.01	0.01	0.01		1.00

Carvalho South - 05/19/2023: Corn, silage

Carvalho South Corn

Sample and source description: Carvalho South Corn

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,000.00	1,700.00	20,400.00		5.60
DL	0.01	0.01	0.01		1.00

Dads - 11/05/2022: Oats, silage-soft dough

Vics Field Oats

Sample and source description: Vics Field Oats

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	2,100.00	12,400.00		5.50
DL	0.01	0.01	0.01		1.00

Dads - 06/03/2023: Corn, silage

Dads Field Corn

Sample and source description: Dads Field Corn

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,600.00	2,800.00	14,600.00		4.80
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

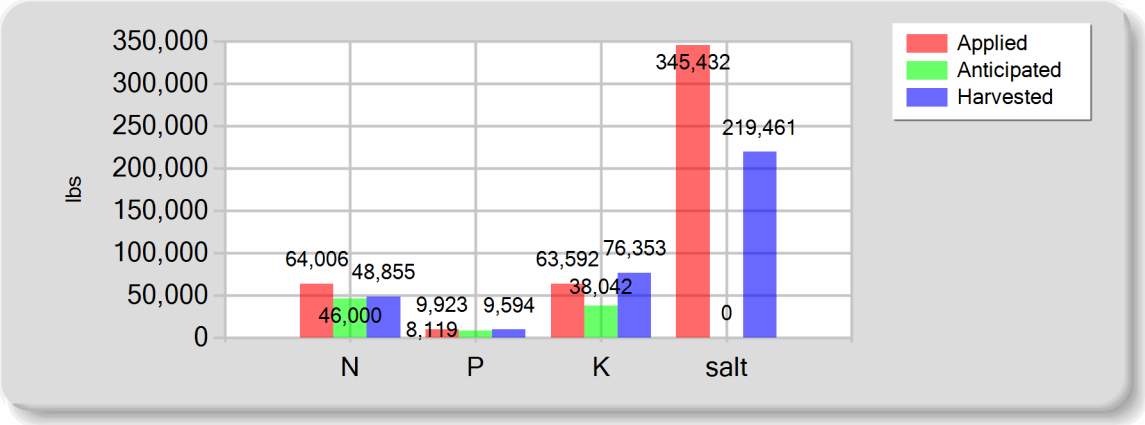
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

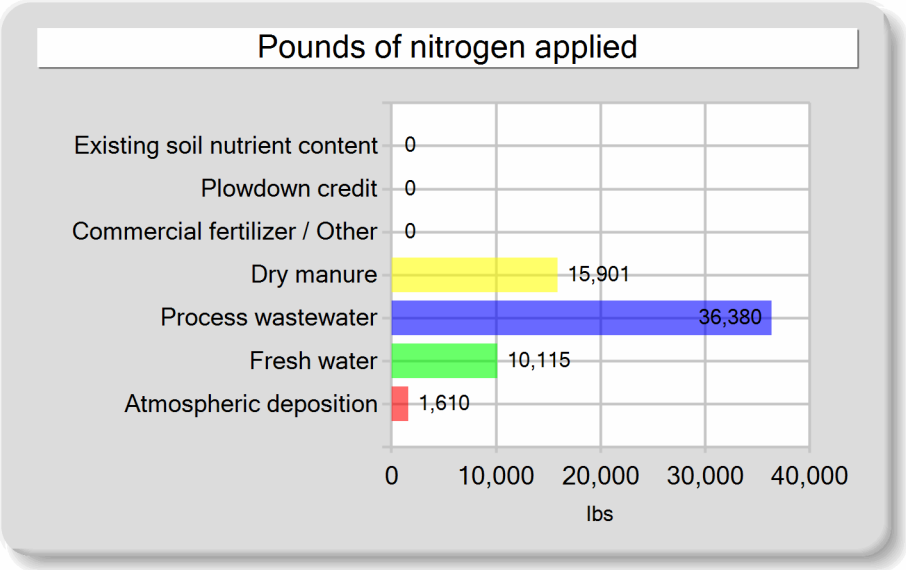
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

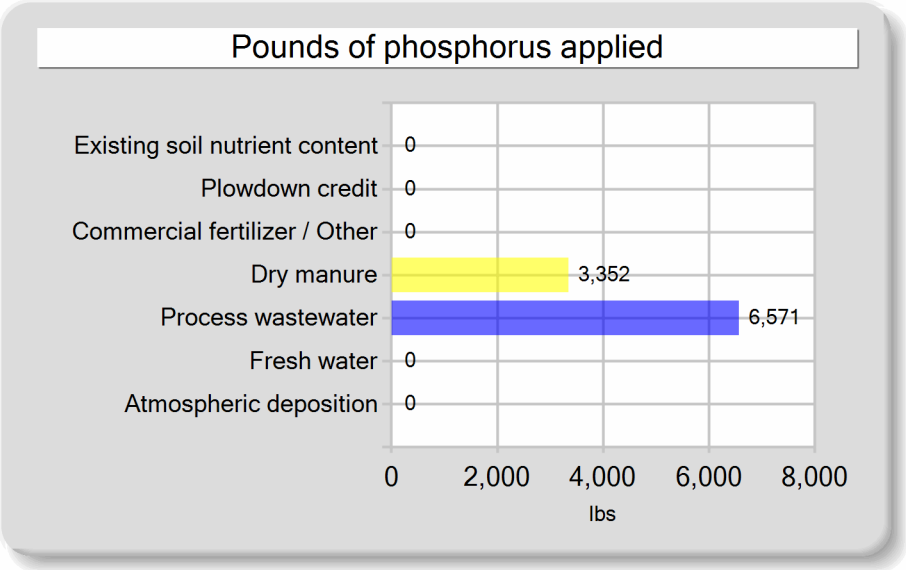
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	15,900.69	3,352.18	4,841.38	94,370.28
Process wastewater	36,380.36	6,571.21	58,750.78	251,061.87
Fresh water	10,114.98	0.00	0.00	0.00
Atmospheric deposition	1,610.00	0.00	0.00	0.00
Total nutrients applied	64,006.03	9,923.38	63,592.16	345,432.15
Anticipated crop nutrient removal	46,000.00	8,119.00	38,042.00	0.00
Actual crop nutrient removal	48,855.11	9,593.66	76,353.00	219,460.92
Nutrient balance	15,150.93	329.72	-12,760.84	125,971.23
Applied to removed ratio	1.31	1.03	0.83	1.57

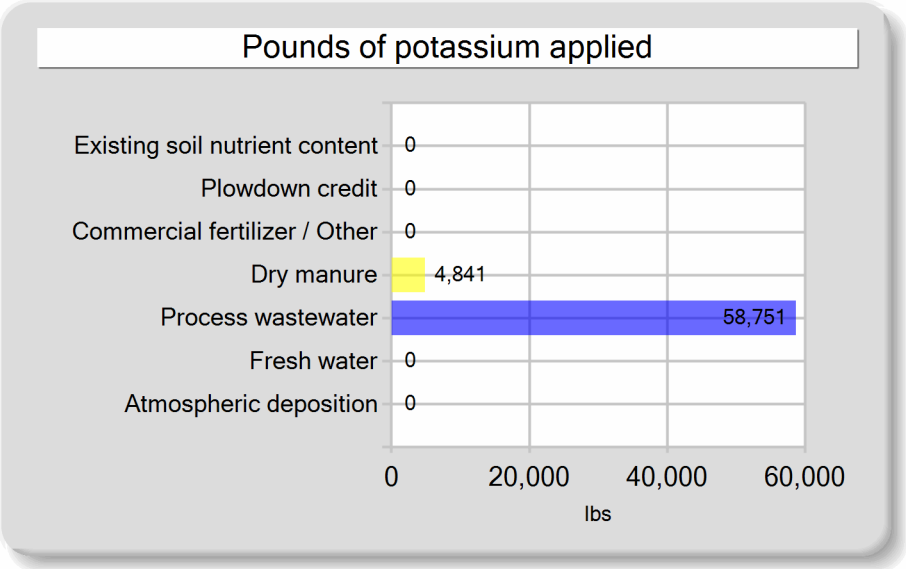
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

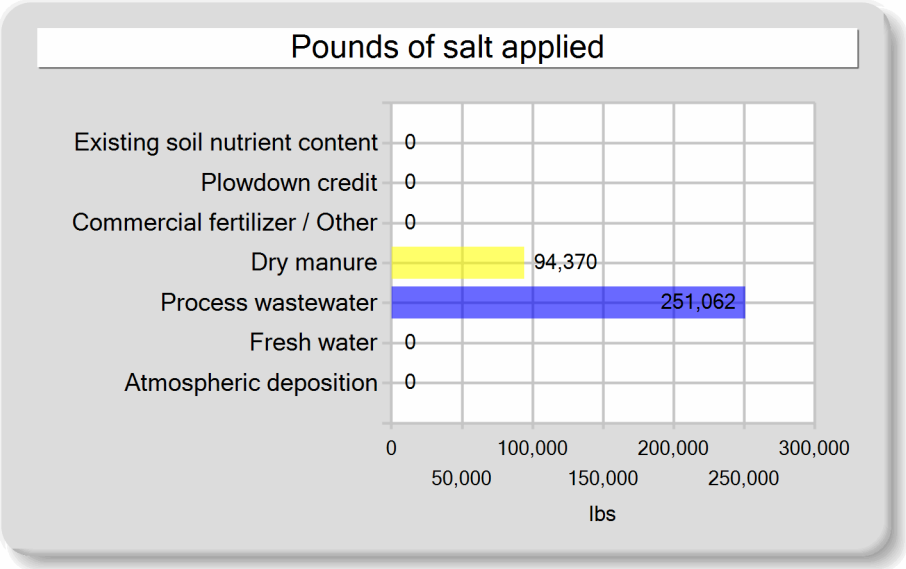


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Field Carvalho South has low N removal rate on oats and corn due to the discharge that occurred under the Owner of that fields blatant application of human sewage to the field. Discharge reports were submitted and clean up was organized with operator and water board, see all discharge information submitted by Fanelli Dairy in regards to this field

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Victor fanelli

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Victor Fanelli	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jul 1, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Victor Fanelli

Name of Dairy Facility: Fanelli Dairy

Facility Address:

7689 Washington RD	Hillmar	Merced	95324
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Victor Fanelli (209) 678-2858
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Silva & Sons LLC

Address of Hauling Company/Person:

23616 Williams AVE	Hilmar	CA	95324
Number and Street	City	State	Zip Code

Contact Person: Frank Silva (209) 632-8202
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Fanelli Dairy - Heifers (209) 678-2858
Name Phone Number

7689 Washington RD	Hilmar	CA	95324
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

7689 Washington RD	Hilmar	95324
Address	City	Zip Code

	Merced
Street and nearest cross street (if no address)	County

Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 01/01/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 288.00 tons

Manure Solids Content: 25.9 %

Method used to determine amount of manure:

tons per load

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Victor fanelli — 1/1/23

Operator Signature — Date

Adam fanelli — 1/1/23

Hauler Signature — Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Victor Fanelli

Name of Dairy Facility: Fanelli Dairy

Facility Address:

7689 Washington RD	Hillmar	Merced	95324
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Victor Fanelli (209) 678-2858
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Silva & Sons LLC

Address of Hauling Company/Person:

23616 Williams AVE	Hilmar	CA	95324
Number and Street	City	State	Zip Code

Contact Person: Frank Silva (209) 632-8202
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Fanelli Dairy - Heifers (209) 678-2858
Name Phone Number

7689 Washington RD	Hilmar	CA	95324
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

7689 Washington RD	Hilmar	95324
Address	City	Zip Code

	Merced
Street and nearest cross street (if no address)	County

Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 04/29/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 670.00 tons

Manure Solids Content: 25.9 %

Method used to determine amount of manure:

tons per load

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Victor fanelli — 4/29/23

Operator Signature — Date

Adam fanelli — 4/29/23

Hauler Signature — Date

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Fanelli Dairy
7689 Washington Rd
Hilmar CA 95324
00-0023474 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0960
Sampled By: M. Pedroso
Requested By: Victor Fanelli
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW-Cow Pens	12/13/23	17:07	1080	1.08	17.7	ND		7.7

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Fanelli Dairy
7689 Washington Rd
Hilmar CA 95324
00-0023474 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0960
Sampled By: M. Pedroso
Requested By: Victor Fanelli
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW-Cow Pens	12/13/23	17:07	**1080**	**1.08**	**17.7**	48.32	4.03

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Fanelli Dairy
7689 Washington Rd
Hilmar, CA 95324

Account# 00-0023474
Account Manager: Ben Nydam
Submitted By: Victor Fanelli

Received: 12/14/2023 14:20
Reported: 12/21/2023 09:52

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0960-01	DW-Cow Pens	Ag Water	M. Pedroso		12/13/2023 17:07

Default Cooler Temperature on Receipt °C: 11.3
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Submitted By: Victor Fanelli

Received: 12/14/2023 14:20
Reported: 12/21/2023 09:52

Sample Results

Sample: DW-Cow Pens
23L0960-01 (Water)

Sampled: 12/13/2023 17:07
Sampled By: M. Pedroso

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.08**	mmhos/cm	0.01	1		12/18/23 16:07	SM 2510 B		BEL0700
Electrical Conductivity umhos	**1080**	umhos/cm	10.0	1		12/18/23 16:07	SM 2510 B		BEL0700
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 17:07	Field		BEL0623
Nitrate Nitrogen as NO3N	**17.7**	mg/L	0.1	1	10	12/15/23 20:31	EPA 300.0		BEL0597
Temperature	**25.0**	units	0.0	1		12/18/23 16:07	SM 4500-H+	H	BEL0700
pH	**7.7**	units	1.0	1		12/18/23 16:07	SM 4500-H+	H	BEL0700

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Account# 00-0023474
Account Manager: Ben Nydam
Submitted By: Victor Fanelli

Received: 12/14/2023 14:20
Reported: 12/21/2023 09:52

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0597										
Blank (BEL0597-BLK1)					Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0597-BLK2)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0597-BLK3)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0597-BLK4)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0597-BLK5)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0597-BS1)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		99.6	90-110		
LCS (BEL0597-BS2)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		103	90-110		
LCS (BEL0597-BS3)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
LCS (BEL0597-BS4)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L	5.000		107	90-110		
Duplicate (BEL0597-DUP1)		**Source: 23L0943-02**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	4.4		0.1	mg/L		4.4			1.52	10
Duplicate (BEL0597-DUP2)		**Source: 23L0943-13**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	2.3		0.1	mg/L		2.2			1.96	10
Duplicate (BEL0597-DUP3)		**Source: 23L0954-06**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	1.6		0.1	mg/L		1.6			1.76	10
Duplicate (BEL0597-DUP4)		**Source: 23L0957-03**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	3.5		0.1	mg/L		3.5			1.20	10
Matrix Spike (BEL0597-MS1)		**Source: 23L0943-02**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	10.2		0.1	mg/L	5.000	4.4	116	90-110		
Matrix Spike (BEL0597-MS2)		**Source: 23L0943-13**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	7.3		0.1	mg/L	5.000	2.2	102	90-110		

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LABORATORY INC

Fanelli Dairy
7689 Washington Rd
Hilmar, CA 95324

Account# 00-0023474
Account Manager: Ben Nydam
Submitted By: Victor Fanelli

Received: 12/14/2023 14:20
Reported: 12/21/2023 09:52

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0597 (Continued)										
Matrix Spike (BEL0597-MS3)			**Source: 23L0954-06**		Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	6.4		0.1	mg/L	5.000	1.6	96.7	90-110		
Matrix Spike (BEL0597-MS4)			**Source: 23L0957-03**		Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	8.7		0.1	mg/L	5.000	3.5	104	90-110		
Reference (BEL0597-SRM1)					Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		99.2	90-110		
Reference (BEL0597-SRM2)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	10.0			mg/L	10.00		100	90-110		
Reference (BEL0597-SRM3)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		
Reference (BEL0597-SRM4)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		
Reference (BEL0597-SRM5)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		

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Hilmar, CA 95324

Account# 00-0023474
Account Manager: Ben Nydam
Submitted By: Victor Fanelli

Received: 12/14/2023 14:20
Reported: 12/21/2023 09:52

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0700										
Blank (BEL0700-BLK1)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.6		1.0	units						
Blank (BEL0700-BLK2)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.6		1.0	units						
Blank (BEL0700-BLK3)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	7.7		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEL0700-DUP1)		**Source: 23L0964-01**			Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1.15		0.01	mmhos/cm		1.15			0.495	10
pH	7.4		1.0	units		7.4			0.135	10
Electrical Conductivity umhos	1150		10.0	umhos/cm		1150			0.495	10
Duplicate (BEL0700-DUP2)		**Source: 23L0965-02**			Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1.13		0.01	mmhos/cm		1.13			0.203	10
Electrical Conductivity umhos	1130		10.0	umhos/cm		1130			0.203	10
pH	7.7		1.0	units		7.7			0.129	10
Reference (BEL0700-SRM1)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	428			umhos/cm	426.0		100	90-110		
Reference (BEL0700-SRM2)					Prepared & Analyzed: 12/18/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0700-SRM3)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEL0700-SRM4)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEL0700-SRM5)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		

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DELLAVALLE™
LABORATORY INC

Fanelli Dairy
7689 Washington Rd
Hilmar, CA 95324

Account# 00-0023474
Account Manager: Ben Nydam
Submitted By: Victor Fanelli

Received: 12/14/2023 14:20
Reported: 12/21/2023 09:52

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0700 (Continued)										
Reference (BEL0700-SRM5)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEL0700-SRM6)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0700-SRM7)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0700-SRM8)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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12/14/23 14:20 23L0960

JG

WATER WORK REQUEST

Bill To: Acct No. 23474 Cons. 8

Purchase Order No. _____ Results Needed By _____

Client: Fanelli Dairy
Address: 7689 Washington Road
City, State, Zip: Hilmar CA 95324
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Victor Fanelli - (209) 668-8097

Facility: _____

Date sampled: 12/13

Sampled by: M Pedroso

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

No. of Samples 1 No. Bottles 1

Water Type: [ ] Drinking [ ] Wastewater
[ ] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

(X) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other _____

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW-Cow pens	Sampled From:	12/13	5:07	Ø	11.3
2.	Sampled From:				
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	M. Pedroso	CASINC	12/13/23 @ 507pm	12/14/23 12p
Second	CSM	DLI	12/14/23 12pm	12/14/23 2:15pm
Third				
Fourth	[illegible]	DLI	12/14 14:20	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____

Shipping
$ _____ In
$ _____ Out
Date _____

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☒ Box ☐ None ☐ **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic	/									
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

12/14/23 14:20 23L0960

SEND OUT

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

Fanelli Dairy
7689 Washington Rd
Hilmar CA 95324
00-0023474 08

E-mail:
Copy To: ;cas.labs@yahoo.com

Lab No.: 23L0734
Sampled By: M Dekeyrel
Requested By: Victor Fanelli
Submitted Date: 12/12/23
Reported Date: 12/18/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW - MB	12/11/23	11:40	1080	1.08	15.3	ND		7.9

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Fanelli Dairy
7689 Washington Rd
Hilmar CA 95324
00-0023474 08

E-mail:
Copy To: ;cas.labs@yahoo.com

Lab No.: 23L0734
Sampled By: M Dekeyrel
Requested By: Victor Fanelli
Submitted Date: 12/12/23
Reported Date: 12/18/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW - MB	12/11/23	11:40	**1080**	**1.08**	**15.3**	41.77	3.48

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
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LABORATORY INC

Fanelli Dairy
7689 Washington Rd
Hilmar, CA 95324

Account# 00-0023474
Account Manager: Ben Nydam
Submitted By: Victor Fanelli

Received: 12/12/2023 16:25
Reported: 12/21/2023 08:59

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0734-01	DW - MB	Well Water	M Dekeyrel		12/11/2023 11:40

Default Cooler Temperature on Receipt °C: 0.3
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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7689 Washington Rd
Hilmar, CA 95324

Account# 00-0023474
Account Manager: Ben Nydam
Submitted By: Victor Fanelli

Received: 12/12/2023 16:25
Reported: 12/21/2023 08:59

Sample Results

Sample: DW - MB
23L0734-01 (Water)

Sampled: 12/11/2023 11:40
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.08**	mmhos/cm	0.01	1		12/13/23 19:00	SM 2510 B		BEL0587
Electrical Conductivity umhos	**1080**	umhos/cm	10.0	1		12/13/23 19:00	SM 2510 B		BEL0587
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 11:30	Field		BEL0405
Nitrate Nitrogen as NO3N	**15.3**	mg/L	0.1	1	10	12/13/23 09:23	EPA 300.0		BEL0350
Temperature	**25.0**	units	0.0	1		12/13/23 19:00	SM 4500-H+	H	BEL0587
pH	**7.9**	units	1.0	1		12/13/23 19:00	SM 4500-H+	H	BEL0587

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DELLAVALLE™ LABORATORY INC

Fanelli Dairy
7689 Washington Rd
Hilmar, CA 95324

Account# 00-0023474
Account Manager: Ben Nydam
Submitted By: Victor Fanelli

Received: 12/12/2023 16:25
Reported: 12/21/2023 08:59

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0350										
Blank (BEL0350-BLK1)					Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0350-BLK2)					Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0350-BLK3)					Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0350-BLK4)					Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0350-BLK5)					Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0350-BS1)					Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	4.6		0.1	mg/L	5.000		92.8	90-110		
LCS (BEL0350-BS2)					Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	5.4		0.1	mg/L	5.000		107	90-110		
LCS (BEL0350-BS3)					Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000		94.3	90-110		
LCS (BEL0350-BS4)					Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	4.6		0.1	mg/L	5.000		92.7	90-110		
Duplicate (BEL0350-DUP1)		**Source: 23L0592-01**			Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	0.08		0.1	mg/L		0.08			0.00	10
Duplicate (BEL0350-DUP2)		**Source: 23L0594-01**			Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	0.09		0.1	mg/L		0.09			5.65	10
Duplicate (BEL0350-DUP3)		**Source: 23L0724-01**			Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	0.6		0.1	mg/L		0.6			1.23	10
Duplicate (BEL0350-DUP4)		**Source: 23L0731-01**			Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	0.2		0.1	mg/L		0.2			0.466	10
Matrix Spike (BEL0350-MS1)		**Source: 23L0592-01**			Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	4.6		0.1	mg/L	5.000	0.08	90.7	90-110		
Matrix Spike (BEL0350-MS2)		**Source: 23L0594-01**			Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000	0.09	91.7	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0350 (Continued)										
Matrix Spike (BEL0350-MS4)			**Source: 23L0731-01**		Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000	0.2	93.7	90-110		
Reference (BEL0350-SRM1)					Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	9.2			mg/L	10.00		92.0	90-110		
Reference (BEL0350-SRM2)					Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	9.3			mg/L	10.00		92.6	90-110		
Reference (BEL0350-SRM3)					Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	9.3			mg/L	10.00		92.7	90-110		
Reference (BEL0350-SRM4)					Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	9.4			mg/L	10.00		93.6	90-110		
Reference (BEL0350-SRM5)					Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	9.2			mg/L	10.00		92.2	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0587										
Blank (BEL0587-BLK1)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	5.5		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEL0587-BLK2)					Prepared & Analyzed: 12/13/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	7.3		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEL0587-BLK3)					Prepared & Analyzed: 12/13/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	7.7		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEL0587-DUP1)		**Source: 23L0731-02**			Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	0.34		0.01	mmhos/cm		0.33			0.509	10
pH	7.2		1.0	units		7.3			1.66	10
Electrical Conductivity umhos	335		10.0	umhos/cm		333			0.509	10
Duplicate (BEL0587-DUP2)		**Source: 23L0737-03**			Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	0.68		0.01	mmhos/cm		0.66			3.31	10
Electrical Conductivity umhos	682		10.0	umhos/cm		659			3.31	10
pH	8.3		1.0	units		8.3			0.00	10
Reference (BEL0587-SRM1)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	448			umhos/cm	426.0		105	90-110		
Reference (BEL0587-SRM2)					Prepared & Analyzed: 12/13/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0587-SRM3)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	1080			umhos/cm	1000		108	90-110		
Electrical Conductivity umhos	1080			umhos/cm	1000		108	90-110		
Reference (BEL0587-SRM4)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	1070			umhos/cm	1000		107	90-110		
Electrical Conductivity umhos	1070			umhos/cm	1000		107	90-110		
Reference (BEL0587-SRM5)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		

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DELLAVALLE™
LABORATORY INC

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Submitted By: Victor Fanelli

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0587 (Continued)										
Reference (BEL0587-SRM5)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEL0587-SRM6)					Prepared & Analyzed: 12/13/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0587-SRM7)					Prepared & Analyzed: 12/13/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0587-SRM8)					Prepared & Analyzed: 12/13/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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12/12/23 16:25 23L0734

WATER WORK REQUEST

Bill To: Acct No. 23474 Cons. 8

Purchase Order No. Results Needed By

Client: Fanelli Dairy
Address: 7689 Washington Road
City, State, Zip: Hilmar CA 95324
Phone Fax
Cell/Email

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Victor Fanelli - (209) 668-8097

Facility:

Date sampled: 12/11/23

Sampled by: mdekeuvel

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

No. of Samples 1 No. Bottles 1

Water Type: [ ] Drinking [ ] Wastewater
[X] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other

Analysis and Bottles Required: *(Please Indicate Analysis)*

(✓) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW-CowPens	Sampled From:	12/11/23	1130A	Ø	~~3~~ JG
~~2. DW-MB~~	Sampled From:	~~12/11/23~~	~~1140A~~	~~Ø~~	0.3
3. not recieved	Sampled From:				
4. MB was recerved	Sampled From:				
5. 50 12/20/23	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	MDekeuvel	CASINC	12/11/23 1140A	12/12/23 @9:30am
Second	JZF	DLV	12-12-23 9:30	12-12-23 2:01
Third				
Fourth		DCI	12.12.23 16:25	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ___ Miles ___ Consulting ___

Amt Paid ___ Rec By ___ Check No. ___

Shipping
$ ___ In
$ ___ Out
Date

Signature

Sample received in cooler with ice?
[ ] Yes [ ] No

rev.update 2020

Shipping Information: Shipped In ☐ Picked-Up ☑ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☑ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☑ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

12/12/23 16:25 23L0734

SEND OUT